

Weekly Research Report — Week 2

MITACS Research Internship

Project: Developing Machine Learning GraphRAG Augmenting Techniques for Clinical Case Description and Patient Journey Follow-Ups

Researcher: Harshit Kapur

Summary of Work Done

This week covered supervisor-directed background reading on GraphRAG query strategies, formulation of the project’s research questions, dataset selection, and identification of target venues. The GitHub repository was populated with the reference literature and the Week 1 reading log.

GraphRAG query strategies

Worked through the DataCamp GraphRAG tutorial as directed. Three retrieval strategies are distinguished:

- **Global search** — map-reduce over community summaries, suited to thematic and synthesis questions
- **Local search** — entity neighbourhood traversal, suited to specific factual lookups
- **DRIFT search** — begins broad, then drills into promising regions; suited to complex multi-hop queries

The tutorial demonstrates these on general text (startup essays). The contribution here is applying and evaluating them on clinical data against medical benchmarks, which the tutorial does not attempt.

Research questions

Four research questions were formulated and mapped to benchmarks:

RQ1 — How does GraphRAG compare to traditional RAG and baseline LLMs in clinical question answering across MedQA, PubMedQA and MedMCQA?

RQ2 — Which query strategy — Global, Local or DRIFT — yields the highest accuracy for different question types (factual lookup, multi-hop reasoning, thematic synthesis)?

RQ3 — How does the structure of a clinical knowledge graph built from WHO and NICE guidelines affect retrieval quality across disease domains?

RQ4 — To what extent does combining GraphRAG guideline retrieval with similar-patient retrieval from MIMIC-IV improve clinical decision support over guideline-only retrieval?

Dataset-to-question mapping

Benchmark	Targets	Rationale
MedQA (USMLE)	RQ1, RQ2	standardised clinical reasoning; Local search
PubMedQA	RQ2, RQ3	evidence-based retrieval; Global search
MedMCQA	RQ2, RQ4	multi-hop relational context; DRIFT search

Comparison of the two core reference papers

Reviewing Sesen et al. (2025) and Chen et al. (2026) side by side showed they use substantially different datasets and evaluation designs — 70 synthetic CKD patients against 36,000 SOAP-structured cases, one guideline against 643. Neither establishes how the approaches compare on standardised benchmarks, which motivates the systematic evaluation proposed here.

Target venues

Three IEEE conferences identified, per supervisor direction to submit the same paper to multiple venues:

Venue	Status
Conference 1	deadline confirmed, earliest of the three
Conference 2	deadline confirmed
Conference 3	deadline not yet announced

The earliest deadline sets the near-term schedule. The MS Word IEEE conference template has been downloaded and the paper skeleton created.

Supervisor direction received

Following this update, direction was given to:

- reformulate the research questions as testable **hypotheses** rather than open questions
- focus the research position on **hybridising vector search with knowledge graphs** — “Strengthening GraphRAG” — and make this one of the stated targets
- maintain experimentation in a Colab directory with IPYNB and dataset URLs accessible
- direct writing effort at the draft itself rather than weekly reports, with progress summarised in the covering email

These are carried into Week 3.

Plan for Week 3

1. Reformulate RQ1–RQ4 as hypotheses
2. Draft Introduction and Related Research in the IEEE template
3. Set up the Colab environment and access MedQA, PubMedQA, MedMCQA
4. Run a first baseline-versus-RAG comparison to establish the pipeline

Blockers

- Awaiting confirmation on LLM API budget. \$500 of the \$1,000 publication budget may be allocated to OpenAI API usage, with a receipt required.
- MIMIC-IV credentialing not yet started; decision pending on whether clinical patient data is needed for the benchmark phase.